

PetCare Hub – Online Pet Marketplace and Care Management System

ICT 1209 Mini Project Proposal Pet Marketplace and Veterinary Service Web
Application



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01. Project Overview

1.1 Introduction

PetCare Hub is a web-based pet care and marketplace system developed to provide a convenient platform for users to explore pets, view pet-care products, manage their own pets, add products to a shopping cart, and book appointments with veterinary doctors.

The system also provides a dedicated administrative dashboard through which administrators can manage users, pets, products, categories, doctors, appointments, and contact messages.

The project combines a user-friendly frontend interface with a PHP and MySQL backend to provide dynamic data management and authentication functionality.

1.2 Purpose of the Project

The main purpose of PetCare Hub is to provide a centralized web platform for common pet-related activities.

The system allows users to:

- Browse available pets.
- Search and filter pets by type.
- View pet-care products and prices.
- Add products to a shopping cart.
- Manage their own pets.
- View veterinary doctors.
- Book veterinary appointments.
- Send contact messages.
- Access a personalized user dashboard.

Administrators can manage the information and activities of the system through a dedicated admin dashboard.

1.3 Target Audience

The main target audience of PetCare Hub includes:

- Existing pet owners.
- People interested in purchasing pets.
- Customers looking for pet-care products.
- Users who need veterinary services.
- Administrators responsible for managing the system.

The system is designed to provide a simple and accessible experience for users who need multiple pet-related services in one platform.

1.4 Key Features

User Features

- User Registration and Login -
Users can create an account and securely log in to the system.
- Pet Marketplace -
Users can browse available pets and view information such as name, type, breed, age, gender and price.
- Pet Search and Filtering -
Users can search for pets and filter them by categories such as dogs, cats, birds, fish, rabbits and other types.
- Product Browsing -
Users can view available pet-care products, their categories, prices and quantities.
- Shopping Cart -
Users can add products to their cart, change quantities, remove products and view the total price.
- My Pets -
Logged-in users can add and manage their own pets through the dashboard.

- **Veterinary Doctors -**
Users can view available veterinary doctors, their specializations, experience and availability.
- **Appointment Booking -**
Users can select their own pet, select a doctor, choose a date and time, and provide a reason for the appointment.
- **Contact Form -**
Users can send messages to the PetCare Hub through the contact form.
- **User Dashboard -**
The dashboard provides information about the user's pets, appointments and cart.

Administrative Features

- **Admin Dashboard -**
Administrators can view summary information about customers, pets, products, doctors, appointments and messages.
- **User Management -**
Administrators can view registered customer accounts.
- **Pet Management -**
Administrators can add, edit and delete pet records.
- **Product Management -**
Administrators can add, edit and delete product records.
- **Category Management -**
Administrators can create, update and delete product categories.
- **Doctor Management -**
Administrators can add, update and delete veterinary doctor information.
- **Appointment Management -**
Administrators can view appointments and update appointment status.
- **Message Management -**
Administrators can view and delete contact messages

02. System Design

2.2 System Architecture

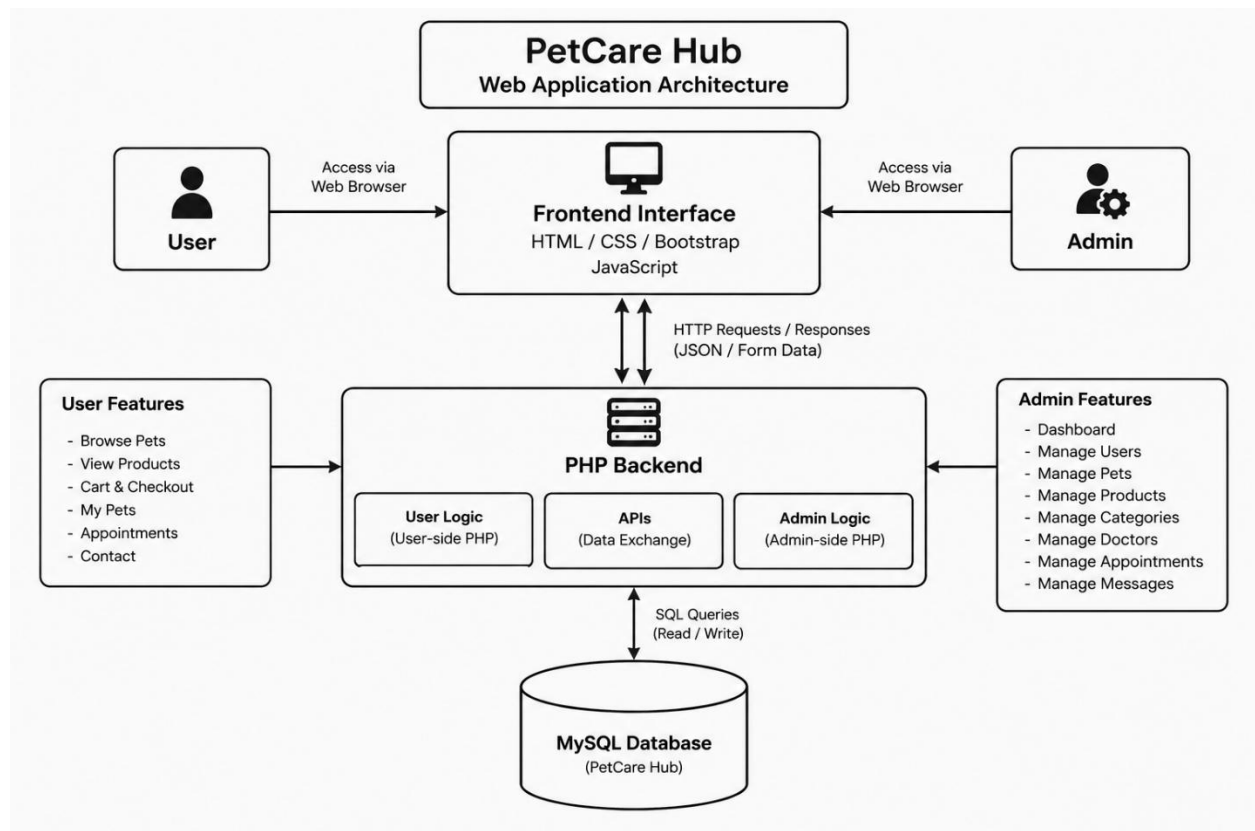


Figure 1 - System Architecture Diagram

2.2 Database Schema

The PetCare Hub system uses a MySQL relational database.

The main tables are:

Table	Main Purpose
users	Stores registered customers and administrator accounts
pets	Stores pet information
products	Stores pet-care products
cart_items	Stores products added to users' carts
doctors	Stores veterinary doctor information
appointments	Stores veterinary appointments
messages	Stores contact messages
categories	Stores product category information

Main Relationships:

A user can own multiple pets and can have multiple appointments and cart items. Each appointment is connected to a user, a pet and a doctor. Cart items connect users with products.

Database Screenshot:

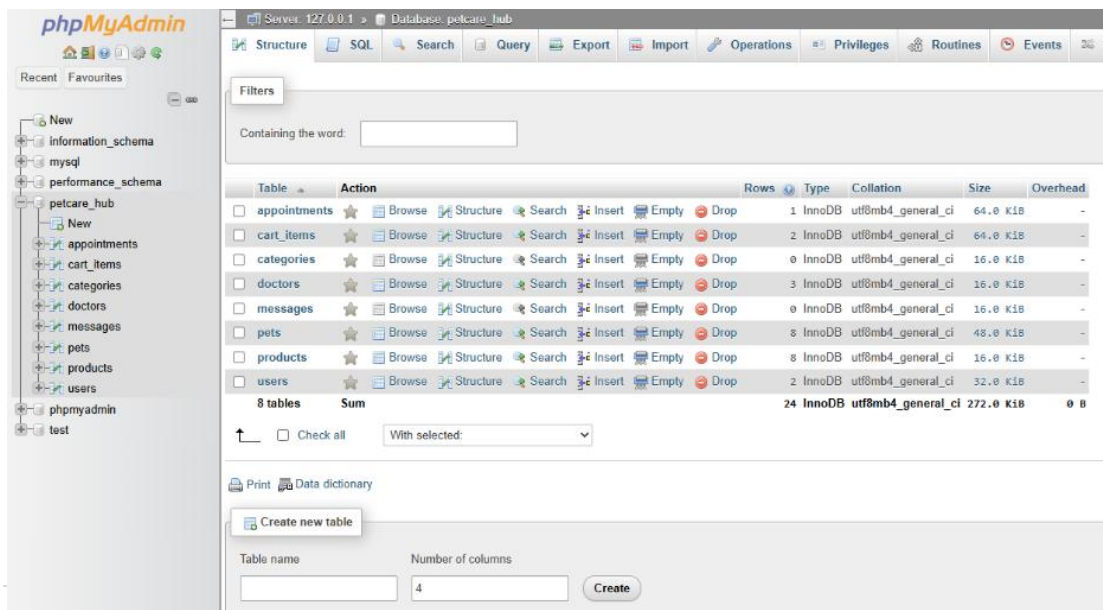
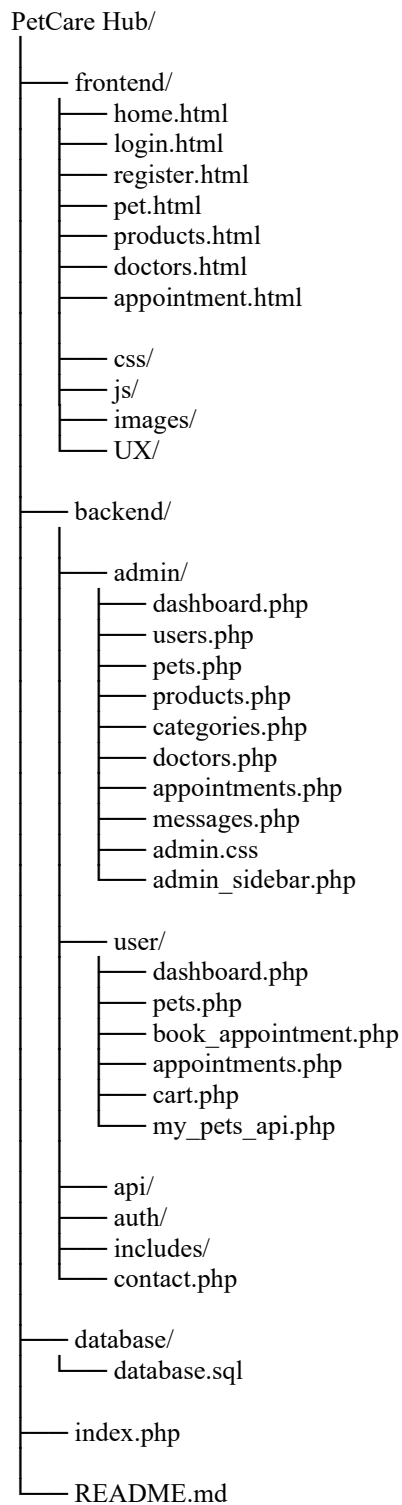


Figure 2 - PetCare Hub MySQL Database Structure

2.3 Folder Structure

The project is organized into separate frontend and backend components.



Folder Structure Explanation

- The frontend folder contains the HTML pages, CSS stylesheets, JavaScript files and image assets.
- The backend/user folder contains user-side PHP functionality, while backend/admin contains administrator functions.
- The backend/api folder contains PHP API endpoints used for communication between the frontend and backend.
- The backend/auth folder contains authentication-related functionality such as login, registration and logout.
- The backend/includes folder contains shared database connection and helper functions.
- The database folder contains the SQL file required to create the project database.

Folder Structure Screenshot

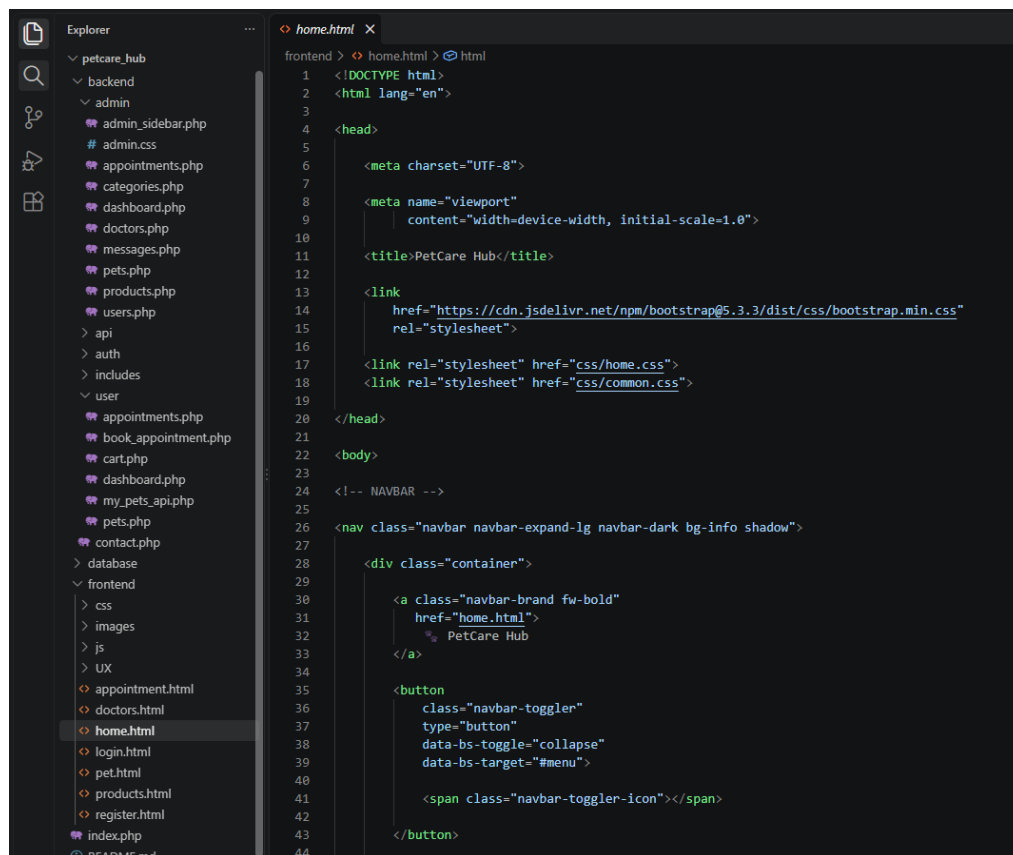


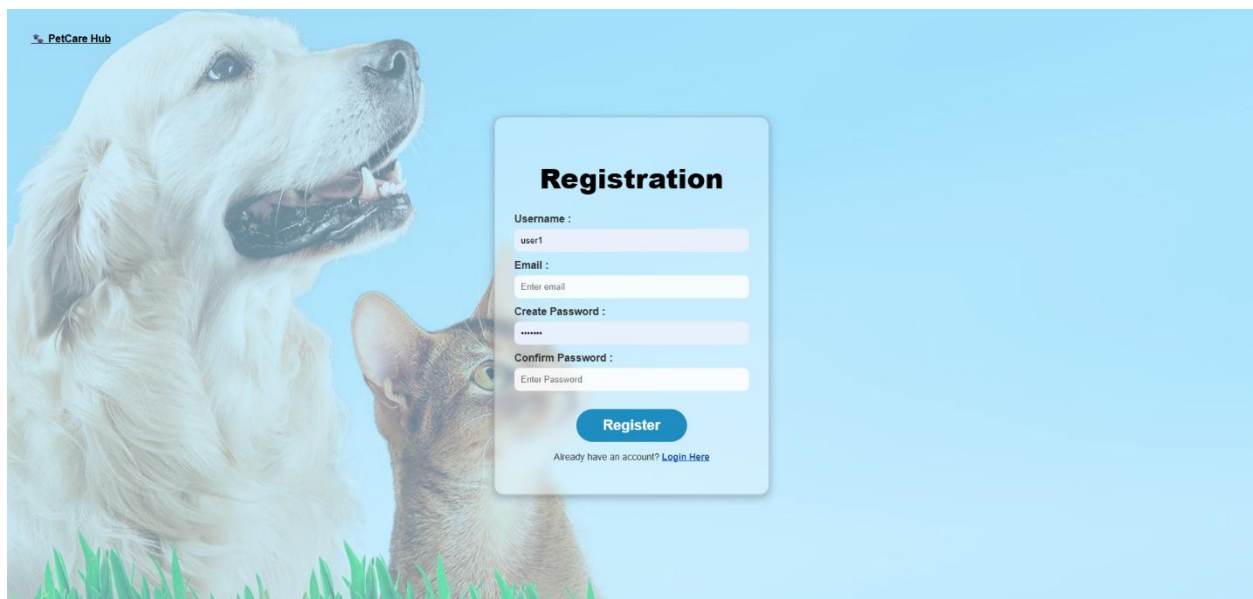
Figure 3 - PetCare Hub Project Folder Structure

03. Implementation

3.1 User Registration

The registration feature allows new users to create PetCare Hub accounts by providing a username, email address and password.

Passwords are processed using PHP password hashing before being stored in the database.



PetCare Hub

Registration

Username :
user1

Email :
Enter email

Create Password :

Confirm Password :
Enter Password

[Register](#)

Already have an account? [Login Here](#)

Figure 4 - User Registration Page

3.2 User Login

The login system allows registered users to log in using their username and password.

After successful authentication, a session is created for the user. The system also distinguishes between customer and administrator accounts and redirects users to the appropriate area.

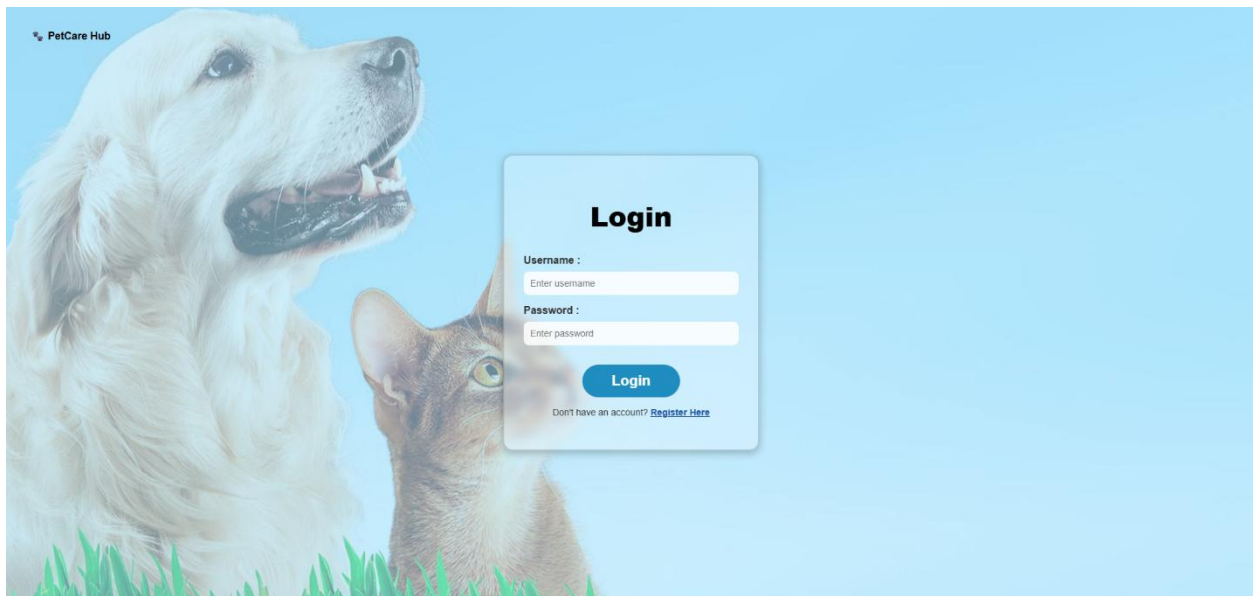


Figure 5 - Login Page

3.3 Home Page

The Home page introduces the PetCare Hub system and provides access to the major services.

- The page contains:
 - PetCare Hub introduction.
 - Featured pets.
 - Featured products.
 - Navigation links.
 - Contact information.
 - Access to the main system features.

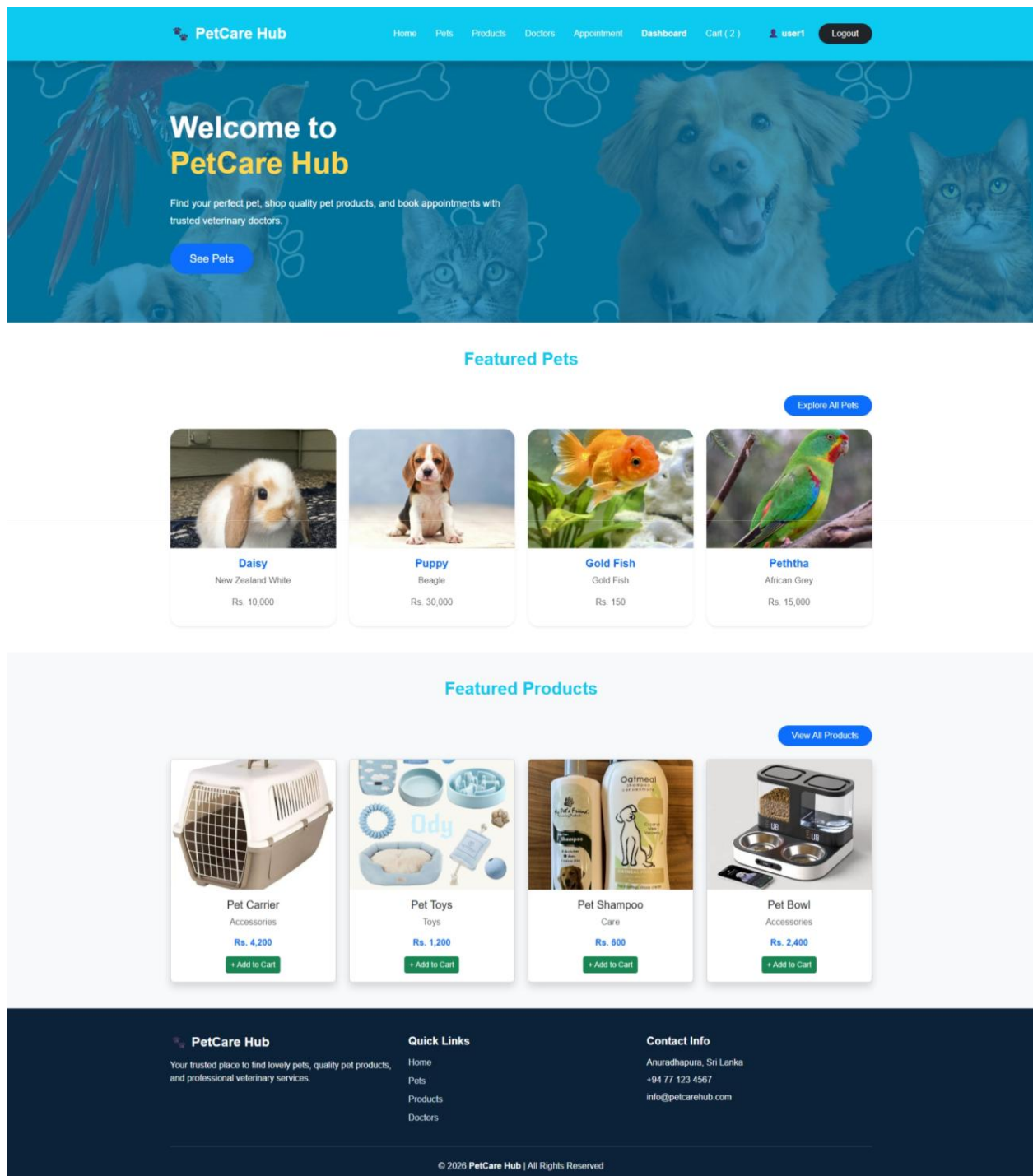


Figure 6 - Home Page

3.4 Pet Marketplace

The Pets page displays available pets dynamically from the database.

- Users can:
 - View available pets.
 - Search for pets.
 - Filter by pet type.
 - View pet information.
 - Navigate through multiple pages of pet records.

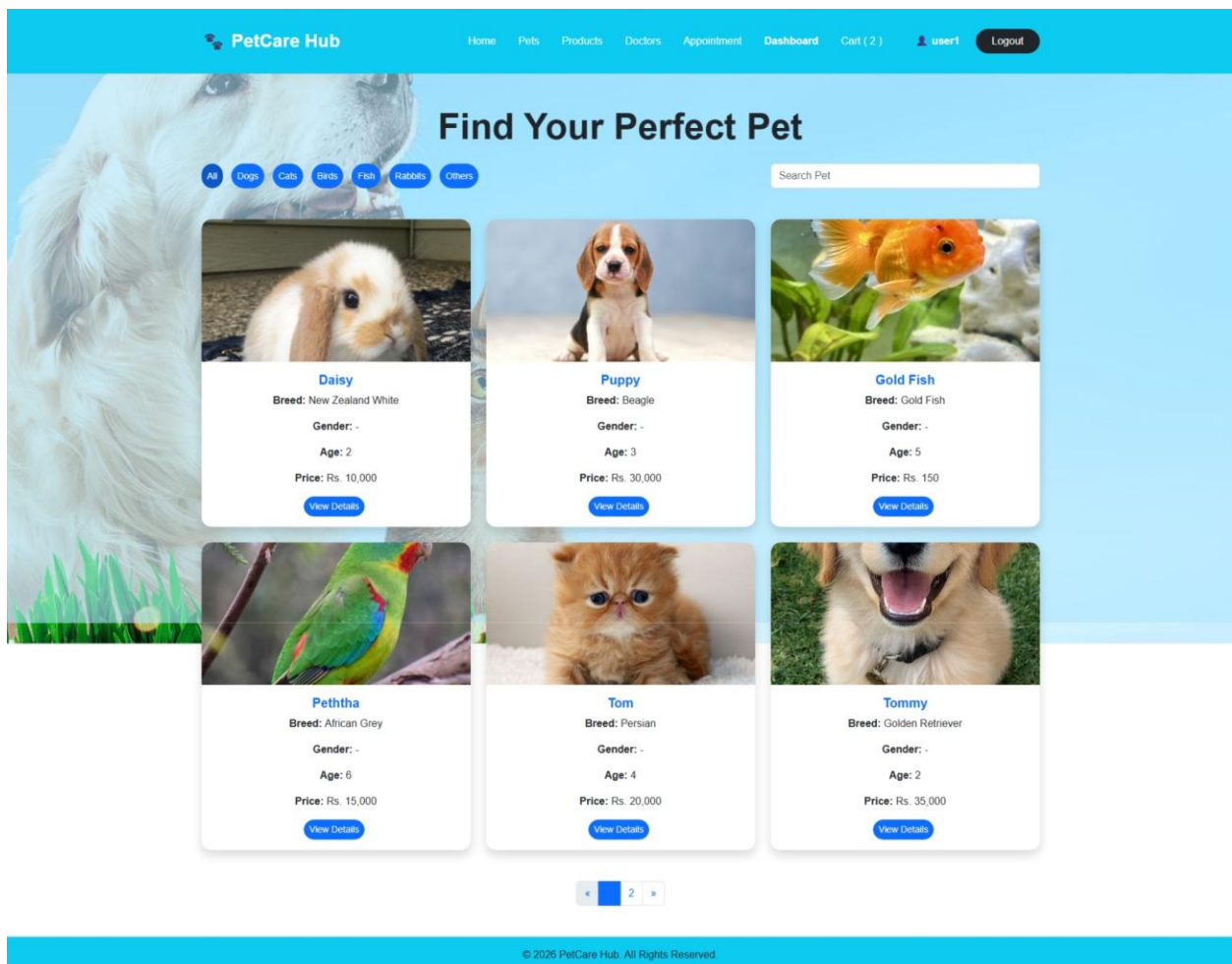


Figure 7 - Pet Marketplace

3.5 Product Browsing

The Products page displays pet-care products stored in the database.

Users can search products and view information such as product name, category and price.

- The system also provides an **Add to Cart** function.

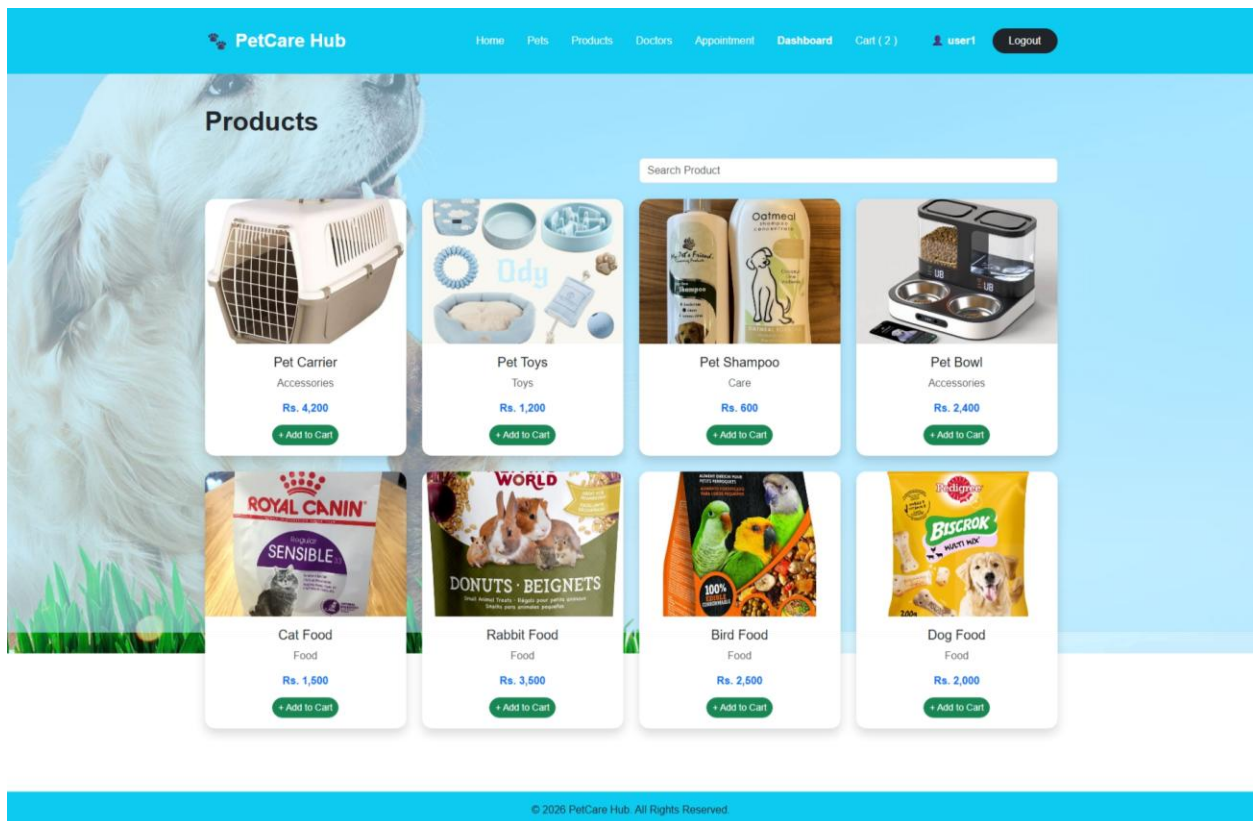


Figure 8 - Products Page

3.6 Shopping Cart

The shopping cart allows users to manage selected products.

- Users can:
 - Add products.
 - Change quantities.
 - Remove products.
 - Clear the cart.
 - View the total amount.
- The system stores cart information in the `cart_items` database table.

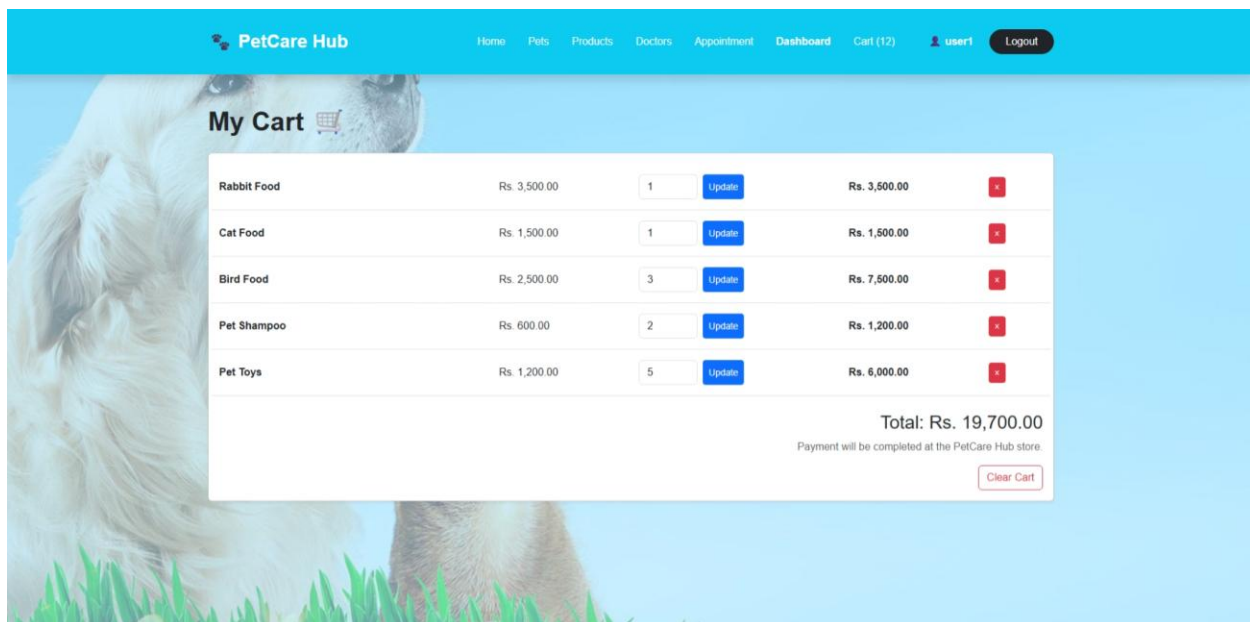


Figure 9 - Shopping Cart

3.7 User Dashboard

The User Dashboard provides a personalized overview of the user's account.

- It displays:
 - Number of pets.
 - Number of appointments.
 - Number of cart items.

It also provides quick links to commonly used functions.

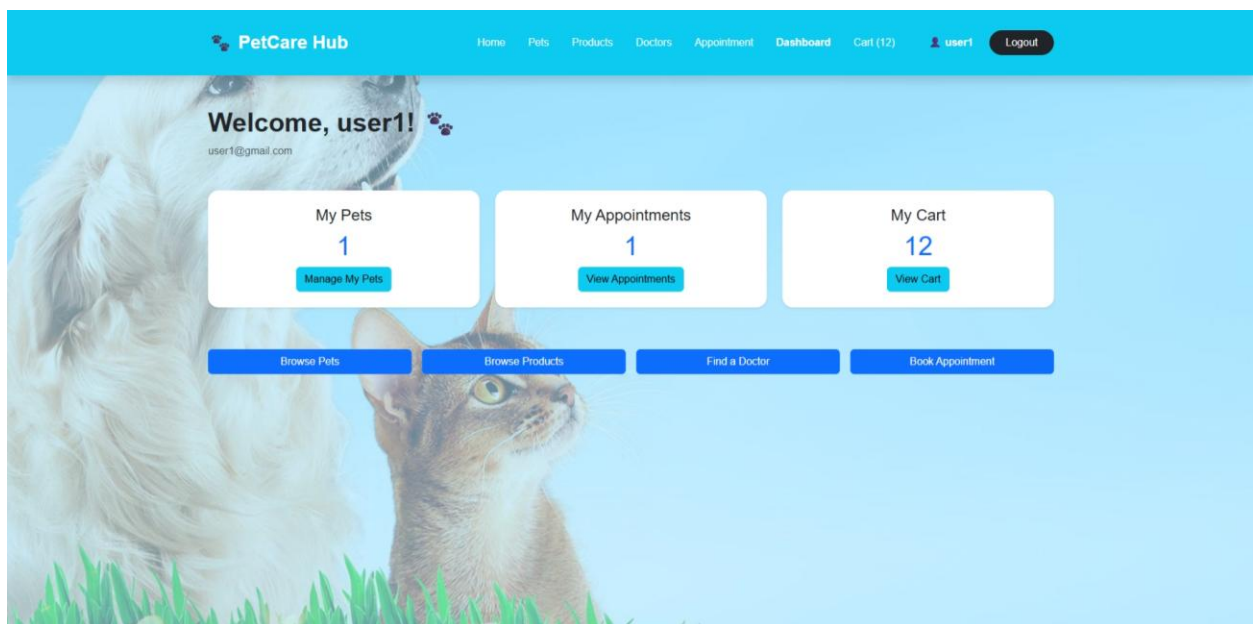
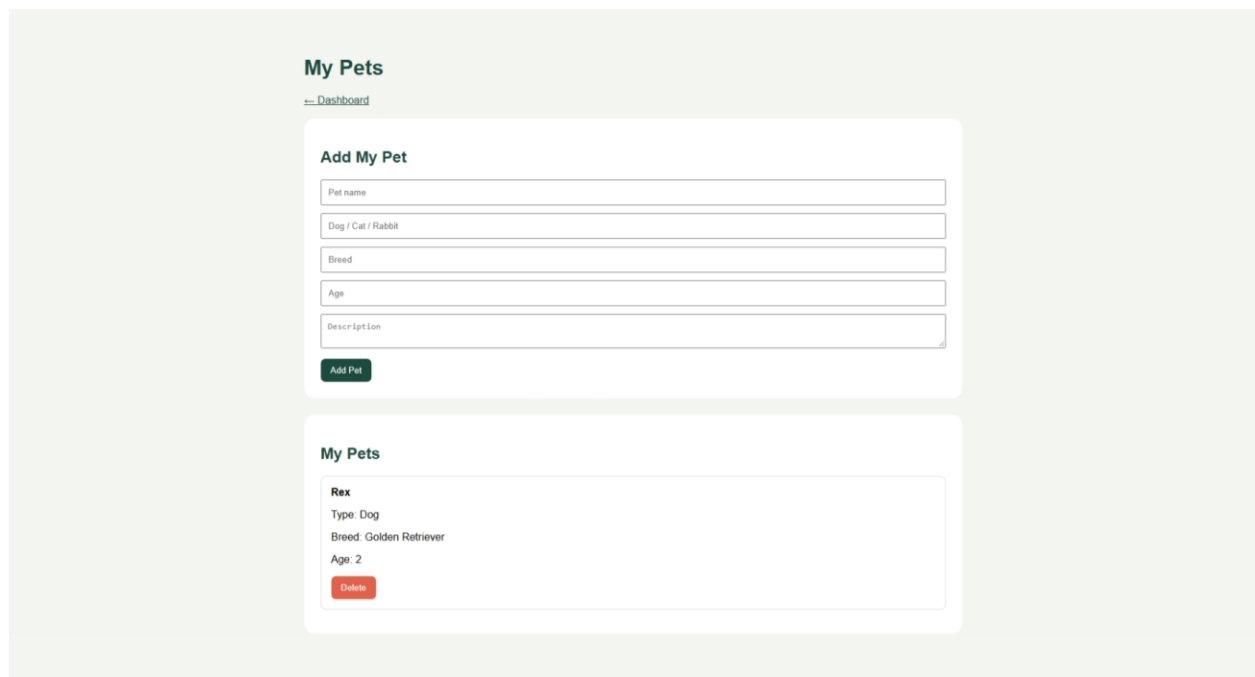


Figure 10 - User Dashboard

3.8 My Pets

The My Pets feature allows logged-in users to add and manage information about their own pets.

- Pet records are associated with the logged-in user through the owner_id field.
- This information is also used by the appointment booking system.



The screenshot displays the 'My Pets' management interface. At the top, there is a header 'My Pets' with a back arrow and a link to 'Dashboard'. Below this is a form titled 'Add My Pet' with the following fields: 'Pet name', 'Dog / Cat / Rabbit', 'Breed', 'Age', and 'Description'. An 'Add Pet' button is located at the bottom of the form. Below the form is a section titled 'My Pets' containing a list of existing pets. The first pet listed is 'Rex', a Dog, Golden Retriever, 2 years old. An 'Delete' button is visible next to the pet entry.

Pet Name	Type	Breed	Age
Rex	Dog	Golden Retriever	2

Figure 11 - My Pets Management

3.9 Veterinary Doctors

The Doctors page retrieves doctor information from the backend database.

- Users can view:
 - Doctor name.
 - Specialization.
 - Experience.
 - Availability.
 - Appointment booking option.

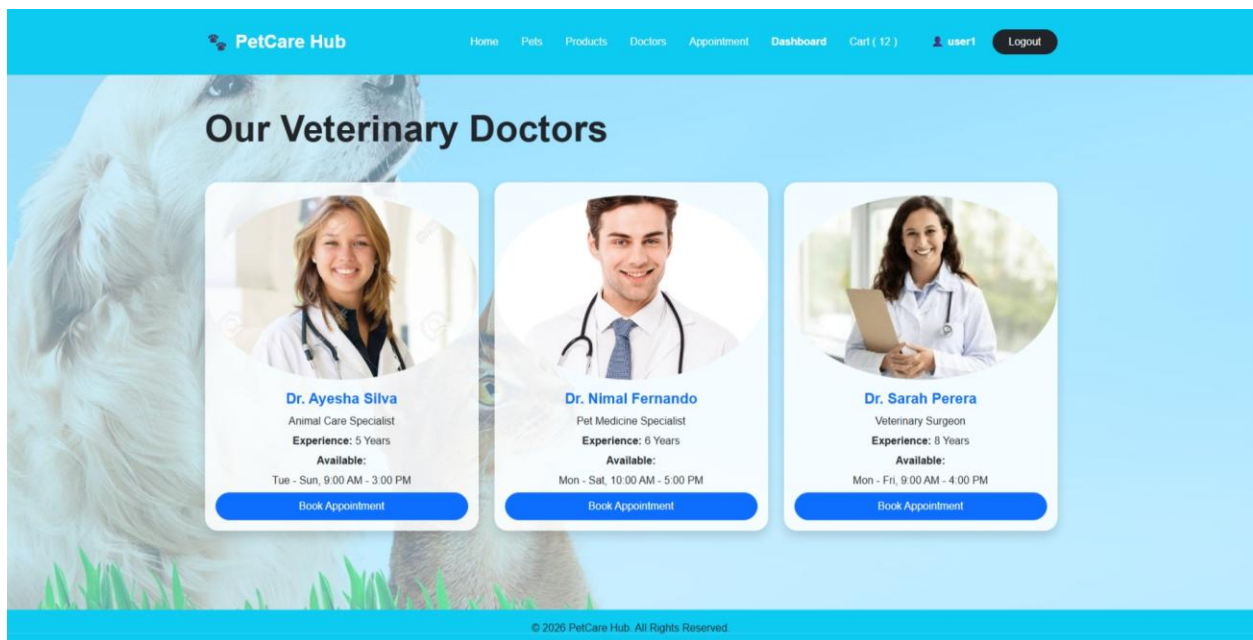
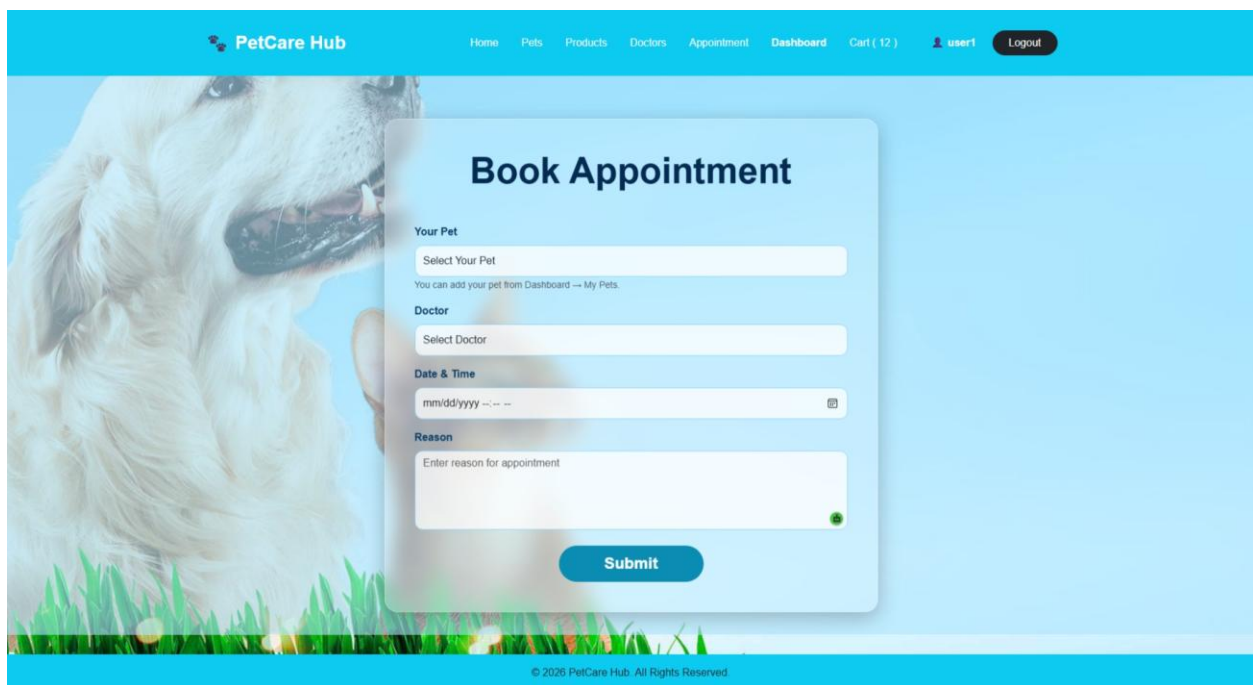


Figure 12 - Veterinary Doctors

3.10 Appointment Booking

The Appointment feature allows users to book veterinary appointments.

- The form includes:
 - User's pet.
 - Veterinary doctor.
 - Appointment date and time.
 - Reason for the appointment.
- The selected information is sent to the PHP backend and stored in the appointments table.



The screenshot shows the 'Book Appointment' form on the PetCare Hub website. The form is overlaid on a background image of a white dog. The form has a title 'Book Appointment' and four main sections: 'Your Pet', 'Doctor', 'Date & Time', and 'Reason'. Each section has a text input field. The 'Your Pet' section has a dropdown menu for 'Select Your Pet' and a note: 'You can add your pet from Dashboard → My Pets.' The 'Doctor' section has a dropdown menu for 'Select Doctor'. The 'Date & Time' section has a date and time picker. The 'Reason' section has a text area for 'Enter reason for appointment'. A green 'Submit' button is at the bottom of the form. The website header includes the PetCare Hub logo and navigation links: Home, Pets, Products, Doctors, Appointment, Dashboard, Cart (12), user1, and Logout. The footer includes the copyright notice: © 2026 PetCare Hub. All Rights Reserved.

Figure 13 - Appointment Booking

3.12 Admin Dashboard

The administrative dashboard provides an overview of the system.

- It displays counts for:
 - Customers.
 - Pets.
 - Products.
 - Doctors.
 - Appointments.
 - Messages.
- A fixed sidebar provides access to all administrative functions.

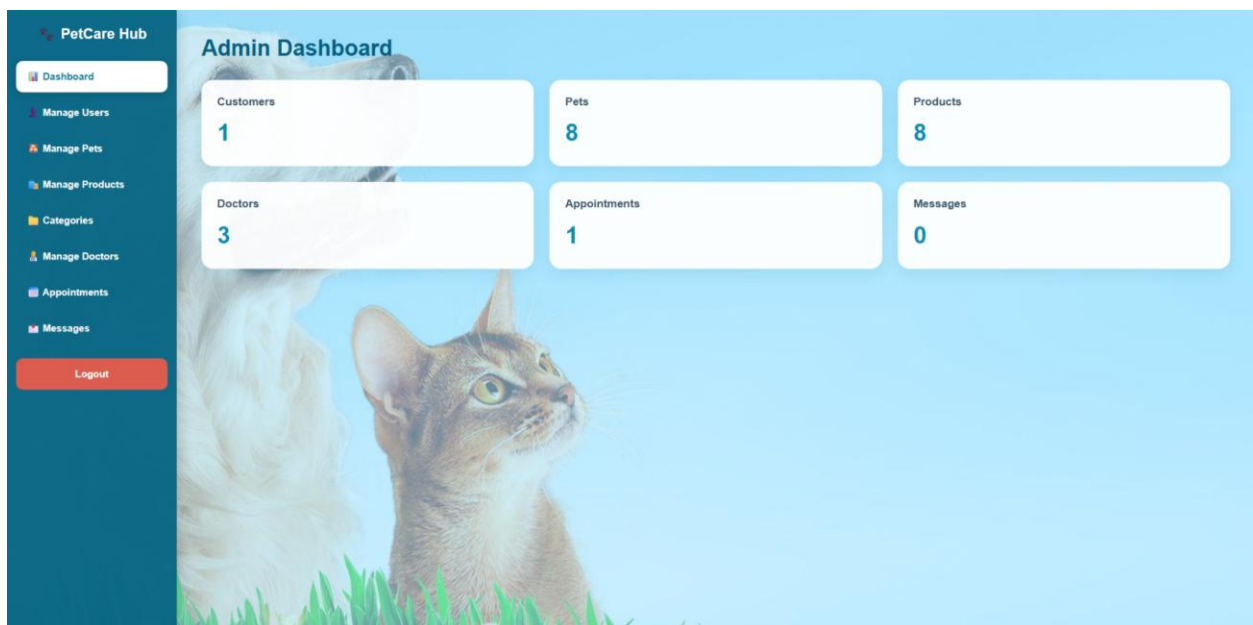


Figure 14 - Admin Dashboard

3.13 Manage Users

The Manage Users page displays registered customer accounts.

- Administrator accounts are excluded from the customer list, while the system continues to maintain administrator access internally.

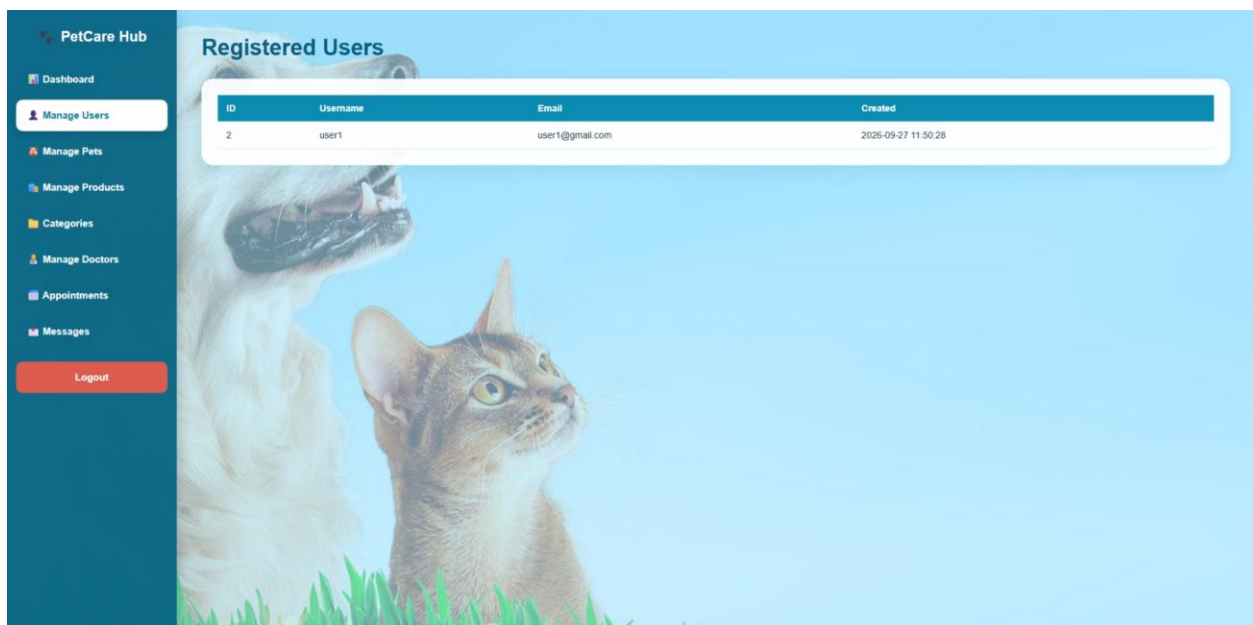


Figure 15 - Manage Users

3.14 Manage Pets

The Manage Pets page allows administrators to:

- Add pets.
- Edit pet details.
- Delete pets.
- View pet owners.

Pet Management

Add New Pet

Pet Name:

Type:

Breed:

Age:

Price:

Image Filename:

Description:

All Pets

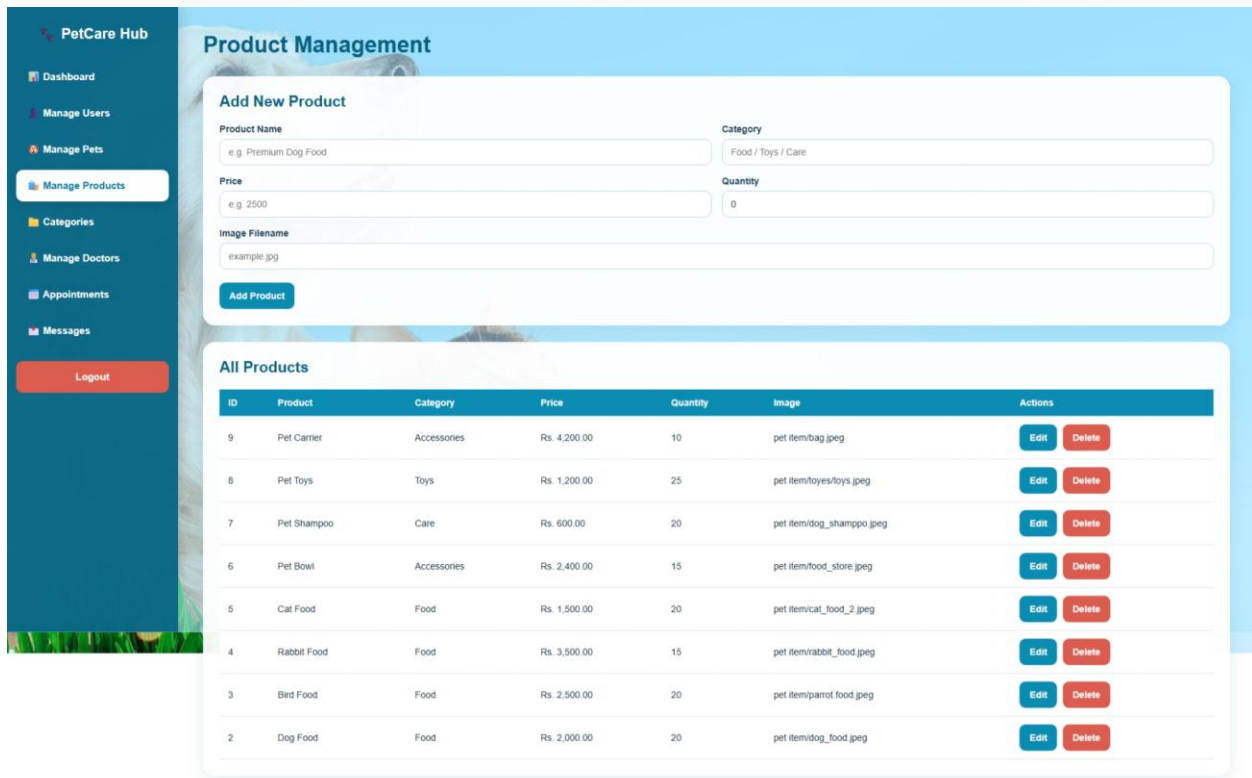
ID	Name	Type	Breed	Age	Price	Owner	Actions
12	Rex	Dog	Golden Retriever	2	Rs. 0.00	user1	Edit Delete
8	Daisy	Rabbit	New Zealand White	2	Rs. 10,000.00	admin	Edit Delete
7	Puppy	Dog	Beagle	3	Rs. 30,000.00	admin	Edit Delete

Figure 16 - Manage Pets

3.15 Manage Products

The Manage Products page allows administrators to:

- Add products.
- Edit products.
- Delete products.
- Update prices and quantities.



PetCare Hub

Product Management

Add New Product

Product Name:

Category:

Price:

Quantity:

Image Filename:

Add Product

All Products

ID	Product	Category	Price	Quantity	Image	Actions
9	Pet Carrier	Accessories	Rs. 4,200.00	10	pet item/bag.jpeg	Edit Delete
8	Pet Toys	Toys	Rs. 1,200.00	25	pet item/toyes/toys.jpeg	Edit Delete
7	Pet Shampoo	Care	Rs. 600.00	20	pet item/dog_shampoo.jpeg	Edit Delete
6	Pet Bowl	Accessories	Rs. 2,400.00	15	pet item/food_store.jpeg	Edit Delete
5	Cat Food	Food	Rs. 1,500.00	20	pet item/cat_food_2.jpeg	Edit Delete
4	Rabbit Food	Food	Rs. 3,500.00	15	pet item/rabbit_food.jpeg	Edit Delete
3	Bird Food	Food	Rs. 2,500.00	20	pet item/parrot.food.jpeg	Edit Delete
2	Dog Food	Food	Rs. 2,000.00	20	pet item/dog_food.jpeg	Edit Delete

Figure 17 - Manage Products

3.16 Manage Categories

The Categories page allows administrators to add, edit and delete product categories.

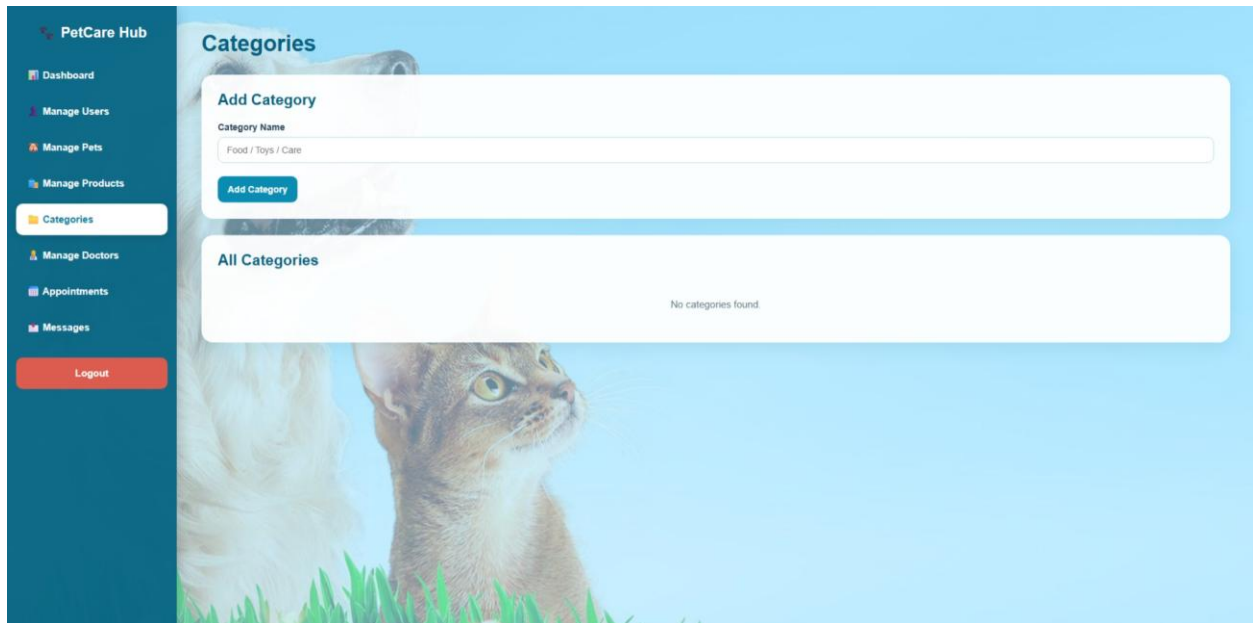


Figure 18 - Category Management

3.17 Manage Doctors

The Manage Doctors page allows administrators to add and edit doctor details and remove doctors from the system.

PetCare Hub

Doctor Management

Add Doctor

Name: Dr. John Silva

Specialization: Veterinary Surgeon

Experience: 8 Years

Email: doctor@example.com

Phone: 0712345678

Image Filename: doctor1.jpg

Availability: Mon - Fri, 9 AM - 4 PM

Add Doctor

All Doctors

ID	Name	Specialization	Experience	Phone	Availability	Actions
3	Dr. Ayesha Silva	Animal Care Specialist	5 Years	0774567890	Tue - Sun, 9:00 AM - 3:00 PM	Edit Delete
2	Dr. Nimal Fernando	Pet Medicine Specialist	6 Years	0723456789	Mon - Sat, 10:00 AM - 5:00 PM	Edit Delete
1	Dr. Sarah Pereira	Veterinary Surgeon	8 Years	0712345678	Mon - Fri, 9:00 AM - 4:00 PM	Edit Delete

Figure 19 - Doctor Management

3.18 Manage Appointments

Administrators can view all appointments and update their status.

- Available appointment states include:
 - Pending
 - Approved
 - Completed
 - Cancelled

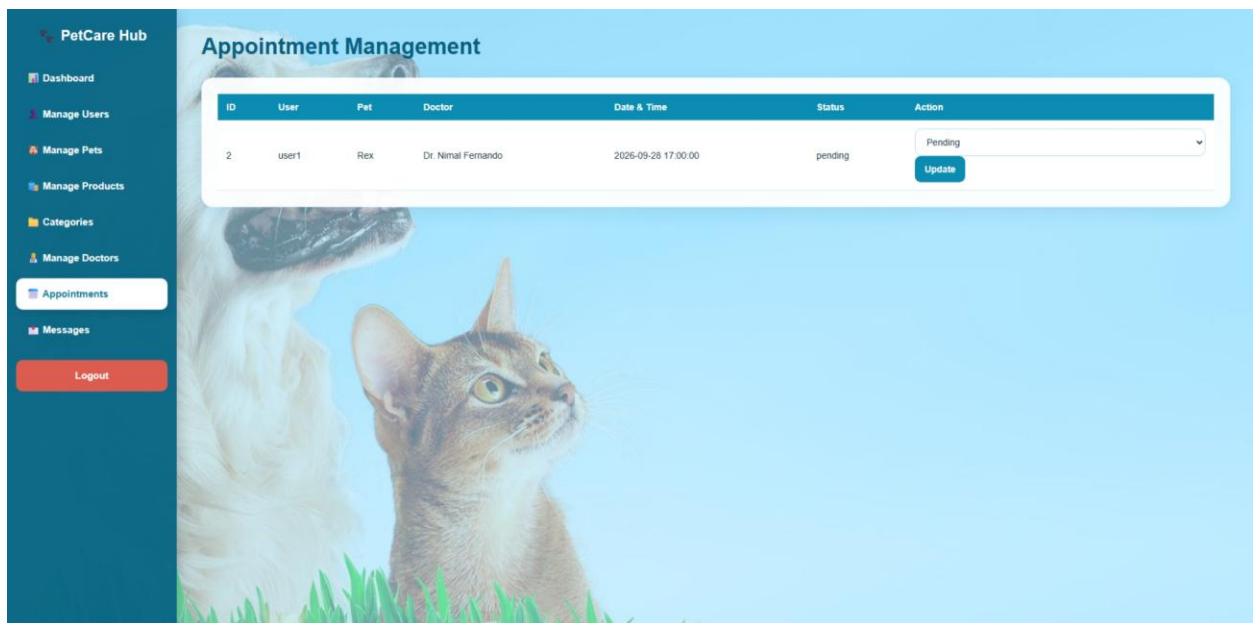


Figure 20 - Appointment Management

3.19 Manage Messages

Administrators can view contact messages submitted by users and remove messages when necessary.

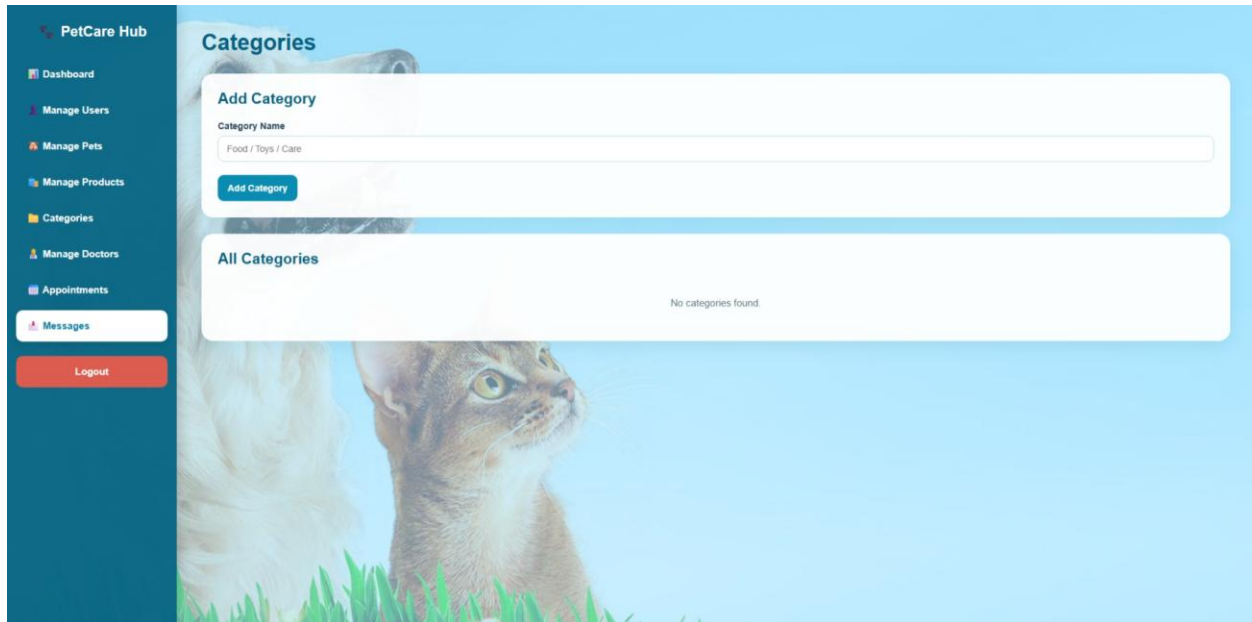


Figure 21 - Message Management

4. Challenges and Solutions

4.1 Frontend and Backend Integration

One challenge was connecting the static frontend pages with the dynamic PHP backend.

The frontend needed to communicate with PHP API files and receive database information dynamically.

- **Solution:**

JavaScript fetch() requests were used to communicate with PHP endpoints. JSON responses were used for dynamic information such as pets, products and doctors.

4.2 Authentication and Session Management

The system required different access levels for ordinary users and administrators.

- **Solution:**

PHP sessions were used to maintain logged-in users. Authentication functions were implemented to restrict access to user and administrator pages according to the user's role.

4.3 User-Specific Pet Information

The appointment system needed to show only pets belonging to the currently logged-in user.

- **Solution:**

Pet records were associated with users using owner_id. A backend request retrieves pets belonging to the current user and displays them dynamically in the appointment form.

4.4 Consistent Admin Interface

Different administrative pages initially contained different sidebar sizes, font sizes, colors and spacing.

- **Solution:**

A shared admin.css stylesheet and reusable admin_sidebar.php component were introduced. This created a consistent sidebar width, typography, spacing, buttons and background across the admin section. Active-page highlighting was also implemented.

4.5 API and File Path Errors

During development, some frontend API requests initially used incorrect file paths or endpoint names.

- **Solution:**

The API paths were checked against the actual PHP filenames and corrected. Separate error handling was also added to make API failures easier to identify.

4.6 Pet Image Path Issue

During development, a problem occurred with newly added pet images. The pet record could be created successfully, but the corresponding image was not displayed correctly because of image path/file handling.

- **Solution / Current Status:**

The image path and file-handling process were investigated while keeping the database functionality intact. The remaining image display issue is related to the relationship between the stored image filename and the physical image location.

5. Individual Contribution

5.1 Member 1 – [W.U.S.T.BANDARA]

UI/UX Design

Member 1 contributed to the design planning and frontend user experience of the PetCare Hub system, including the structure and visual organization of the main user-facing interfaces.

Frontend Development

Member 1 was primarily responsible for frontend implementation, including:

- HTML page development.
- CSS styling.
- Bootstrap components.
- Responsive layout.
- Navigation structures.
- User interface improvements.
- JavaScript interactions.
- Search and filtering.
- Pagination.
- Dynamic frontend content.

Database Development

Member 1 was responsible for designing and implementing the MySQL database structure and maintaining the SQL database file.

This included database tables and relationships required for:

- Users.
- Pets.
- Products.
- Cart items.
- Doctors.
- Appointments.
- Messages.
- Categories.

User-Side Backend

Member 1 also contributed to the user-side PHP functionality, including:

- User dashboard.
- User pet management.
- Shopping cart interface.
- Appointment-related user functionality.
- Contact message submission.

Testing and Integration

Member 1 also participated in testing, debugging and final system integration.

5.2 Member 2 – [M.N.A.JAYAWEERA]

UI/UX Design

Member 2 contributed to the UI/UX planning and interface design using Figma, helping establish the overall layout and usability structure of the application.

Admin Backend

Member 2 was primarily responsible for the administrative section, including:

- Admin dashboard.
- Manage users.
- Manage pets.
- Manage products.
- Manage categories.
- Manage doctors.
- Manage appointments.
- Manage messages.

API Development

Member 2 contributed to the PHP API layer used to exchange information between the frontend and backend, including data retrieval and submission functionality.

Authentication

Member 2 contributed to:

- Login functionality.
- Registration processing.
- Logout functionality.
- Session management.
- Role-based access.

Shared Backend Functions

Member 2 also contributed to shared PHP functionality such as database connectivity and helper functions.

Testing and Integration

Member 2 participated in testing, debugging, backend integration and final system testing.

6. Setup Instructions

6.1 Required Software

The following software is required:

- XAMPP
- Web browser
- PHP
- MySQL
- phpMyAdmin

6.2 Start XAMPP

Open XAMPP Control Panel and start:

Apache
MySQL

Both services should be running before opening the system.

6.3 Copy the Project

Copy the complete PetCare Hub project folder into the XAMPP htdocs directory.

6.4 Create the Database

Open the browser and go to:

<http://localhost/phpmyadmin>

Create the required PetCare Hub database.

6.5 Import the SQL File

Select the created database and choose:

Import

Select:

database/database.sql

Then click **Import** or **Go**.

This creates the tables required by the system.

6.6 Configure the Database Connection

Open:

backend/includes/db.php

Make sure the database name and MySQL connection settings match the local XAMPP configuration.

- Database name should be petcare_hub

6.7 Run the Project

Open:

[http://localhost/ petcare_hub /frontend/login.html](http://localhost/petcare_hub/frontend/login.html)

Register a new customer account and log in.

Existing user name and passwords are,

User Name : User1
Password : User123

6.8 Access the Admin Dashboard

The administrator can access:

http://localhost/petcare_hub/backend/admin/dashboard.php

The admin account should be used for administrative functions.

Existing admin name and passwords are,

User Name : admin

Password : Admin@123

6.9 System Testing

After installation, the following should be tested:

- Registration.
- Login and logout.
- Pet browsing.
- Product browsing.
- Shopping cart.
- User pet management.
- Appointment booking.
- Doctor management.
- Admin dashboard.
- User management.
- Product management.
- Pet management.
- Category management.
- Appointment status updates.

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